

# NERC

## Task:

The task was to extract all the Name Entities in the given test set and assign the right BIO-labels to each token in the sentences. As mentioned in the NLTK book chapter 7 section 5, NER consists of two sub-tasks: "identifying the boundaries of the NE, and identifying its type", so that is what we did in this task.

## Data Selection:

In this task we used the CoNLL-2003 dataset to train and validate our NER model. This dataset is often used for training NER systems and it was previously introduced in the lab sessions. The dataset contains sentences with tokens, part-of-speech tag, chunk tags and named entity labels in BIO format. The testset contains sentences with entity names such as personal names, actor names, place names, movies names, etc.

## Pre-Processing:

During the (minimalistic) pre-processing each sentence was read and then split into token and corresponding BIO-label. The CoNLL-2003 data set was processed by removing empty lines and filtering out meta data such as the -DOCSTART-. The tokens were extracted together with their corresponding BIO-labels. The same procedure was applied to both the training and the validation data. Additional cleaning of the TSV file was needed, due to inconsistent formatting in some rows, and were therefore skipped over.

Feature extraction was kept simple, by focusing on the lexical properties of the tokens. Each token was represented using its surface form, extended with lowercase representation and capitalization-based features. This choice was made to establish a strong base line without relying on the contextual embeddings.

## Methodology:

The linear Support Vector Machine (LinearSVC) was used as our supervised machine learning approach. Due to its strong baseline for sequence labeling and its efficiency on sparse feature spaces.

Firstly the data was loaded. Instead of using the ConLLCorpusReader like we did in the lab assignments, I read the data directly from the raw files. The main reason for this was that the reader did not return the correct Named Entity labels, but instead returned chunking labels, which were not compatible with the test data.

Then each token was transformed into a feature representation using these lexical features: word entity, lowercase form of the word and the capitalization feature. These features were transformed into numerical vectors using DictVectorizer. Then the model was trained on a training set and a validation set.

## Results:

The CoNLL-2003 training set contained 203,621 tokens, while the validation set only contained 51,362 tokens. The tokens labelled as O (non-entity) were in both datasets the most frequent. This is typical for NER tasks, but this creates a class imbalance.

This class imbalance resulted in an accuracy of 0.94 and a weighted F1-score of 0.93 for the validation set. The location entities (B-Loc) performed the best, with achieved F1-scores of 0.71 and 0.72. The performance for I-tags such as I-PER and I-ORG, were lower as can be seen in the attached validation table.

When the model was evaluated on the test set, the performance of both the accuracy and the weighted F1-score dropped to 0.81 and 0.74. This is likely caused by the different label schemes between the training and test label schemes. The CoNLL-2003 dataset uses labels such as PER, ORG, LOC, and MISC, whereas the test set also contains labels such as PERSON and WORK\_OF\_ART, which were not present during training. As a result, the model was unable to learn these entity types effectively.

```
Train instances: 203621
Valid instances: 51362

Train label distribution:
Counter({'O': 149578, 'B-LOC': 7168, 'B-PER': 6688, 'B-ORG': 6321, 'I-
PER': 6328, 'I-ORG': 3784, 'B-MISC': 3438, 'I-LOC': 1157, 'I-MISC':
1155})

Valid label distribution:
Counter({'O': 42759, 'B-PER': 1842, 'B-LOC': 1887, 'B-ORG': 1341, 'I-
PER': 1387, 'B-MISC': 932, 'I-ORG': 731, 'I-MISC': 346, 'I-LOC': 25
7})
```

|        | precision | recall | f1-score | support |
|--------|-----------|--------|----------|---------|
| B-LOC  | 0.87      | 0.89   | 0.83     | 1887    |
| B-MISC | 0.87      | 0.73   | 0.79     | 922     |
| B-ORG  | 0.78      | 0.65   | 0.71     | 1341    |
| B-PER  | 0.85      | 0.62   | 0.72     | 1842    |
| I-LOC  | 0.72      | 0.78   | 0.71     | 257     |
| I-MISC | 0.79      | 0.58   | 0.65     | 346     |
| I-ORG  | 0.71      | 0.45   | 0.55     | 731     |
| I-PER  | 0.78      | 0.41   | 0.52     | 1387    |
| O      | 0.95      | 1.00   | 0.98     | 42759   |

|               | precision | recall | f1-score | support |
|---------------|-----------|--------|----------|---------|
| B-LOC         | 0.67      | 0.37   | 0.42     | 7       |
| B-MISC        | 0.80      | 0.80   | 0.80     | 0       |
| B-ORG         | 0.40      | 0.67   | 0.50     | 3       |
| B-PER         | 0.80      | 0.80   | 0.80     | 0       |
| B-PERSON      | 0.80      | 0.80   | 0.80     | 11      |
| B-WORK_OF_ART | 0.80      | 0.80   | 0.80     | 9       |
| I-LOC         | 0.80      | 0.80   | 0.80     | 1       |
| I-ORG         | 0.80      | 0.80   | 0.80     | 2       |
| I-PER         | 0.80      | 0.80   | 0.80     | 0       |
| I-PERSON      | 0.80      | 0.80   | 0.80     | 8       |
| I-WORK_OF_ART | 0.80      | 0.80   | 0.80     | 10      |
| O             | 0.81      | 0.19   | 0.42     | 185     |

|              | accuracy |
|--------------|----------|
| macro avg    | 0.81     |
| micro avg    | 0.85     |
| weighted avg | 0.93     |

## Classification report test data

## Conclusion:

The LinearSVC model provided a strong baseline for NER and achieved good results on the validation data. The low performance on the I-tags in the validation set indicate that the model had more difficulty identifying complete multi-token entities. The performance of the model decreased on the test set due to differences between the training and test datasets. Future improvements could include using a dataset with matching entity labels, adding contextual features, or applying more advanced sequence models such as CRFs or transformer-based models.

## Link to code:

<https://colab.research.google.com/drive/1F-eqoGR-bZi-fxNXAHd4Dsms7XbgdaVescrollTo=A4eaNnteeAs5>

## Literature:

NLTK book, chapter 7, sections 5

# Sentiment Analysis

## TASK

The goal of this task is to classify airline-related text into one of three sentiment categories: Positive, Negative, and Neutral.

## Dataset

The Airline Tweets dataset was used for sentiment classification. The dataset consists of airline-related tweets annotated with positive, negative, or neutral sentiment labels. Both sentiment classification systems were trained and evaluated using the same dataset and train/test split in order to ensure a fair comparison.

## Preprocessing

The text data was loaded together with the corresponding sentiment labels. The dataset was divided into training and test sets using an 80/20 split. No manual feature engineering was performed beyond the feature representations used by the models.

## Methodology

Two sentiment classification systems were evaluated.

The first system uses TF-IDF (Term Frequency-Inverse Document Frequency) to transform text into numerical feature vectors. These vectors were used to train a Linear Support Vector Machine (LinearSVC) classifier. This approach serves as a strong traditional machine learning baseline for text classification.

The second system uses RoBERTa, a transformer-based language model that was fine-tuned for sentiment classification. RoBERTa captures contextual information and semantic relationships between words, allowing it to model language more effectively than traditional feature-based approaches.

## Results

| Model        | Accuracy | Macro F1 |
|--------------|----------|----------|
| SVM + TF-IDF | 0.84     | 0.83     |
| RoBERTa      | ?        | ?        |

-INSERT COMPARISON GRAPH-

## Error Analysis

Most classification errors occurred on neutral tweets, where sentiment signals are often weak or ambiguous. The TF-IDF model relies primarily on word frequencies and may therefore struggle with contextual meaning, sarcasm, and subtle sentiment expressions.

RoBERTa .....

## Conclusion

## Work Division:

The section on Topic Analysis was completed by Ben Gratz,

including finding training data, coding, gathering

visualizations, composition of the analysis section, and

implementing the section into the poster.

The secton on NERC was solved by Emma van der Tol,

including coding, gathering visualizations, composition

of the analysis section, and implementing the section into

the poster.

Sentiment Analysis:

Evelina Ravlusevic

- Coding (TF-IDF + SVM implementation)

- Results and error analysis

- Poster preparation and section integration

## INTRODUCTION

# Topic Analysis - BERT

## Task:

Our task was to assign topic labels to the test dataset. This dataset consisted of reviews of either novels, movies, or restaurants.

## Data Selection:

Initially, I attempted to train the BERT model on the 20 newsgroups test dataset, as we had seen in the lab. However, I quickly realized that this wouldn't be an optimal solution due to the domain mismatch. The newsgroups data consists of news articles and classifies into topics such as, "alt.atheism", "comp.graphics", "sci.med", and "sci.space". Meanwhile, we are testing the model's performance against a dataset of reviews of novels, movies, and restaurants. So, training the model on news articles would not translate well in this scenario. Therefore, I am using the Amazon Multi-Domain Reviews Dataset (ACL) from huggingface.co via Johns Hopkins University. This dataset is composed of Amazon reviews of 4 products - books, DVDs, electronics, and kitchen. Although originally designed for sentiment analysis, those categories match our test categories rather well. So, rather than focusing on the sentiment of the reviews, I focused on the product which was being reviewed.

Data Structure and Processing:

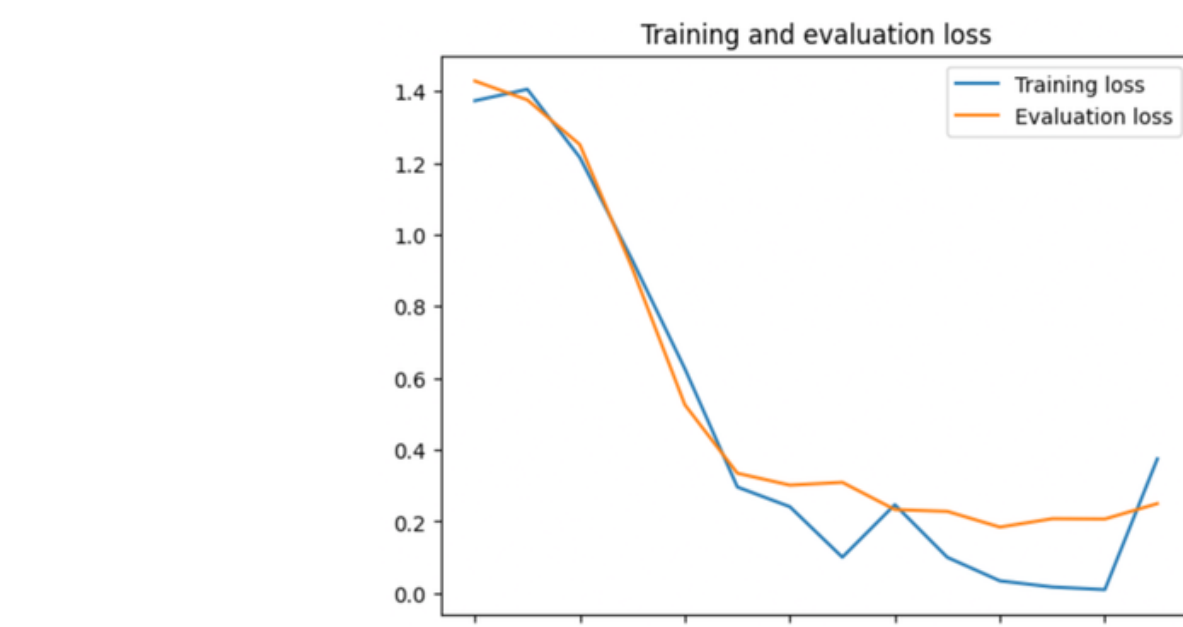
The data was structured into 4 folders, 1 for each of the categories. Inside each of those 4 folders, there are 3 review files, of positive, negative, and unlabeled reviews represented as a bag-of-words. Next, I loaded in a label mapping to convert domain names to numerical values. The list is as follows: "books": 0, " dvd": 1, "kitchen": 2, "electronics": 3. The aim was that, in the test set, books would map to 0, movies would map to 1, and restaurants would map to 2. The dataset is relatively balanced between the 4 categories, but not entirely. There are 7945 reviews for kitchen products (label 2), 7681 for electronics (3), 6465 for books (0), and 5585 for DVDs (1). I applied minimal preprocessing to the training data, as it was already relatively well organized for our purposes. However, I did remove the embedded sentiment labels, "label=positive" and "label=negative", as the model did not cheat or find shortcuts using sentiment labels because this task was only concerned with topic classification. Then, I split the data into training and validation splits with 24909 data points for training and 2768 for validation (90% training, 10% validation), using a stratified split to ensure that the 4 topics are represented in a relatively balanced manner.

## Methodology:

First, I decided to use a fine-tuned BERT model for a few reasons. Primarily, BERT has shown strong performance in topic identification across domains. So, this allows flexibility to work with various training and testing datasets. Next, using a supervised learning approach allows us control over the topics.

After selecting BERT, I had to make some decisions in the fine-tuning process. In "How to Fine-Tune BERT for Text Classification", Sun et al discuss strategies for fine-tuning parameters. When fine-tuning BERT, I used the same learning rate given by an example in this paper, 2e-5. With a low learning rate like this, we avoid the catastrophic forgetting problem. If we had used a more aggressive learning rate, the model may fail to converge. Next, I chose to train the model on 3 epochs given that each epoch has 1557 steps. Many of the other fine-tuning parameters were taken from lab 4.4.

In this graph of training and evaluation loss, we see both drop sharply and together. This indicates that the model learns well and converges. By the end of the training, there is a slight generalization gap, as evaluation loss is higher than training loss. Then, finally, training loss spikes above evaluation loss which could suggest slight overfitting.

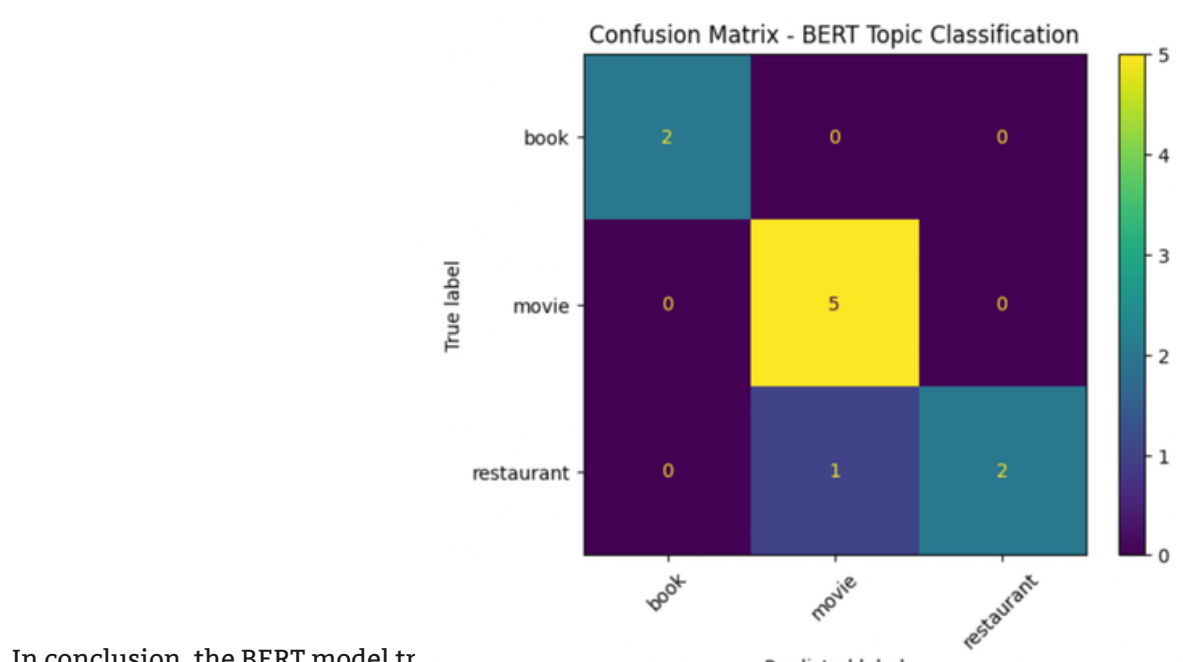


The model achieved an overall accuracy of 0.92 (0.93 precision, 0.92 recall). The recall and precision scores for classes 0 and 2 (books and kitchen) are relatively balanced, but there is a bit of a discrepancy for classes 1 and 3 (DVDs and electronics). For DVDs, the precision of 0.96 and recall of 0.86 suggests that some true instances may be missed. Meanwhile, for electronics, the precision of 0.84 and recall of 0.97 tells us that there are very few false negatives, but more false positives. However, overall, the model scores very highly on the validation data.

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.97      | 0.94   | 0.96     | 646     |
| 1            | 0.96      | 0.86   | 0.90     | 559     |
| 2            | 0.97      | 0.92   | 0.94     | 795     |
| 3            | 0.84      | 0.97   | 0.90     | 768     |
| accuracy     | 0.93      | 0.92   | 0.92     | 2768    |
| macro avg    | 0.93      | 0.92   | 0.92     | 2768    |
| weighted avg | 0.93      | 0.92   | 0.92     | 2768    |

| sentence_id | text | sentiment | topic | predicted_topic | predicted_topic_name |
|-------------|------|-----------|-------|-----------------|----------------------|
| 0           | 0    | 0         | 0     | 0               | books                |
| 1           | 1    | 1         | 1     | 1               | dvd                  |
| 2           | 2    | 2         | 2     | 2               | kitchen              |
| 3           | 3    | 3         | 3     | 3               | electronics          |
| 4           | 4    | 4         | 4     | 4               | electronics          |
| 5           | 5    | 5         | 5     | 5               | electronics          |
| 6           | 6    | 6         | 6     | 6               | electronics          |
| 7           | 7    | 7         | 7     | 7               | electronics          |
| 8           | 8    | 8         | 8     | 8               | electronics          |
| 9           | 9    | 9         | 9     | 9               | electronics          |

As hoped, the model predicts movie reviews into the DVD topic and correctly identifies sentences about books. Furthermore, it correctly predicts 2 out of the 3 sentences about restaurants into the kitchen category, which we chose as our proxy for restaurants. One test sentence was misclassified, and that is a sentence about restaurants being classified into the movie category. A confusion matrix is listed below where we have substituted the labels of "DVD" with "movie" and "kitchen" with "restaurant" to allow for a match between the proxies in our training data and the true labels in the test data.



In conclusion, the BERT model is a well for this task. Although there was a slight mismatch in categories between the testing and training data, we effectively used "kitchen" as a proxy for "restaurant" and "DVD" as a proxy for "movie". Nine out of the ten test sentences were predicted correctly. In the future, however, it could be beneficial to combine datasets to create a training set that more accurately matches the test categories. If we had used a training set that also included restaurant reviews, the one misclassified point would likely be correctly predicted. Finally, as Sun et al. mention, it could be beneficial to further pre-train BERT on in-domain data, such as movie, book, and restaurant reviews, rather than simply the general-domain corpus.

## Links to Code:

[https://colab.research.google.com/drive/1-A0BsmgP7-wlu3v8ZFM1fIqU8\\_Q\\_Phrir1usp-drive\\_link](https://colab.research.google.com/drive/1-A0BsmgP7-wlu3v8ZFM1fIqU8_Q_Phrir1usp-drive_link)

## Works Cited:

Sun, Chi, et al. How to Fine-Tune BERT for Text

Classification? arXiv, 2019,

<https://arxiv.org/abs/1905.05583>.

## Dataset Links:

<https://www.cs.jhu.edu/~mdredze/datasets/sentiment/>

(select the option: processed\_acl.tar.gz (19M))